

Appendices

Appendix: Pipeline Stage Reference I

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Table 1: Single-project DAG exported from default pipeline.yaml (names shown in topological order). Scripts live under scripts/.

Stage name	Script / method	Primary inputs	Outputs / artefacts	Failure mode
Clean Output Directories	_run_clean_outputs	prior projects/<name>/output/, mirrored output/<name>/ targets	emptied trees	Blocking
Environment Setup	scripts/pipeline/stage_00_setup.py	toolchain probes	scaffold dirs, env exports	Blocking
Infrastructure Tests	scripts/pipeline/stage_01_test.py --infr / a-only --infra-scope pipeline-smoke	tests/infra_tests	coverage + junit-style logs	tolerant ceilings
Project Tests	scripts/pipeline/stage_01_test.py --project-only	projects/<name>/tests/	stage artefacts	blocking by default
Project Analysis	scripts/pipeline/stage_02_analysis.py	thin scripts	figures/, data/, reports	Blocking
Connector Search	scripts/pipeline/stage_08_connector_search.py .yaml	manuscript/config	output/data/connector_search/	opt-in (science tag); skipped if not configured
Provenance Record	scripts/pipeline/stage_09_provenance_record.py --stage "Connector Search"	prior stage outputs	.provenance/graph.json	opt-in (provenance tag); skipped if not configured
PDF Rendering	scripts/pipeline/stage_03_render.py	manuscript/, placeholders	.pdf/.tex bundles	Blocking
Output Validation	scripts/pipeline/stage_04_validate.py	render tree	Markdown + PDF diagnostics JSON	Blocking / downgrade

Appendix: Pipeline Stage Reference II

LLM Scientific Review	scripts/pipeline/stage_06_llm_review.py	resolved manuscript artefacts	textual reviews	Optional skip (allow_skip)
LLM Translations	scripts/pipeline/stage_06_llm_review.py	abstract metadata	multilingual snippets	Optional skip (allow_skip)
Copy Outputs	scripts/pipeline/stage_05_copy.py	validated tree	mirrored output/<name>/...	soft fail logged
Ebook Generation	scripts/pipeline/stage_11_ebook.py	rendered combined markdown	output/ebook/ (EPUB/MOBI/DOCX)	opt-in (ebook tag); soft fail
Metadata	scripts/pipeline/stage_12_metadata.py	manuscript/config .yaml	output/metadata/ (ONIX/JSON/OPF)	opt-in (metadata tag); soft fail
Package Executable	scripts/runner/bundle_executable.py	project tree + outputs	container bundle manifest	opt-in (bundle tag)
Archival Publication	scripts/runner/archival_publication.py	bundle + deliverables	archival deposit manifest	opt-in (archival tag)

Appendix: Pipeline Stage Reference I

`scripts/pipeline/stage_07_executive_report.py` is invoked **outside** this DAG whenever `execute_multi_project.py` aggregates pipelines—supplying cross-project KPI dashboards absent from lone-project checkpoints.